Studies in form

RESEARCH NOTE / DEVELOPMENT / JOURNAL EDITION

Development changes the effect of a disturbance

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We often describe an organism as becoming more stable as it develops, but a smaller response can have several causes. An intervention may rearrange less material in an older body, or the body may respond differently to the change it receives. Flow Lenia lets us separate these questions by making copies of a saved state and comparing an intervention with its untouched continuation.

The initial age comparison used the same local swap fraction at each age and found smaller downstream differences in later bodies. Meanwhile, sibling shapes grown from varied initial conditions became more different from one another. Reduced sensitivity to this disturbance therefore coexisted with increasing variation in form.

A later experiment followed 64 specimens at 19 ages, recording the actual matter rearranged and adjusting the swap fraction to try to match the early response. The matching was incomplete: only 1,880 of 3,648 selected interventions came within 25 per cent of their target. Later bodies still showed smaller downstream differences in the adjusted comparison, with the age contrast pointing the same way in all 16 groups. This strengthens the evidence for changing susceptibility, but it does not cleanly separate the disturbance delivered from the dynamics that follow it.

The proposed information peak did not identify a special onset of resistance. A gradual developmental change and a distinctive transition at that peak are different hypotheses, and the evidence supports the former more clearly.

A companion controller comparison made another useful distinction. Whole-state feedback moved bodies toward a target relative to leaving them alone. Once the strongest comparator correctly included doing nothing, its comparative advantage remained unresolved; at the held-out endpoint 128 steps after release, that advantage was negative. Achieving a target while control is active does not establish superiority over simpler alternatives, or a change that maintains itself after control stops.

The useful lesson is that a body needs to be described in relation to the changes we can make to it. Susceptibility, diversity of form, target achievement, and persistence after release are separate measurements. Keeping them separate makes the developmental result more informative, and gives the next experiment a clearer question.

Read the full articles: Developmental susceptibility and Prediction and control. This edition accompanies the revised web research note.